

FNB APP OF THE YEAR HACKATHON 2026 | SAFER COMMUNITIES & GBV RESPONSE

KHUSELA (SafeLedger)

"To Rise Up." A safety & forensic evidence vault disguised as a calculator.

Zero-trace emergency dispatch, ECTA Section 15 cryptographic evidence sealing (AES-256-GCM + SHA-256), and automated Department of Justice Form 06 (J480) Protection Order drafting for South African GBV survivors.

LIVE HOSTED APP & JUDGE DEMO CREDENTIALS (NO COMPILING REQUIRED)

Live Production URL:	**https://khusela.lumesystems.co.za**
GitHub Repository:	**https://github.com/maansjayden/khusela-safeledger**
Disguise Persona:	**Mzansi Smart-Budget (ZAR & 15% VAT Calculator)**
Unlock Vault PIN:	**Type 2580 then press = (Opens Encrypted Vault)**
Silent Duress SOS PIN:	**Type 9999 then press = (Fires Silent SOS Alert)**
Demo Video (Drive):	**drive.google.com/file/d/1q1eLHnxAo4Q8hbUDiJslv-z30z_qL0Yo**

1. The Problem: Why Existing Safety Apps Fail Domestic Violence Survivors

Backed by SAPS Crime Statistics, Medical Research Council Data, and Magistrate's Court Realities

9,516+ Cases / 90 Days

SAPS Q3 2024/2025 Crime Stats

Over 9,516 domestic violence-related assaults and 1,471 domestic homicides recorded in a single quarter, with 1 in 5 SA women experiencing partner violence (Stats SA).

R28B to R42B / Year

KPMG Too Costly to Ignore Study

Gender-Based Violence costs the South African economy between R28.4 billion and R42.4 billion annually (1% of GDP) in lost productivity, healthcare, and legal costs.

3 Deadly Bottlenecks

Device Surveillance & Court Attrition

Survivors living with an abuser face daily phone inspections, deleted or inadmissible gallery evidence, and an intimidating 12-page DoJ Form 06 (J480) affidavit.

The Three Failure Points Khusela Solves:

Pain Point 1: Coercive Device Surveillance

Installing an overtly branded 'GBV Panic App' or dialing 10111 in the same room as a violent partner triggers immediate physical retaliation when the abuser inspects the phone.

Pain Point 2: Fragile, Inadmissible Evidence

Photos of bruises or voice notes saved in a standard phone gallery are routinely discovered and deleted by the abuser, or challenged in court for lacking tamper-evident timestamps.

Pain Point 3: The 12-Page Form 06 (J480) Barrier

Under acute trauma, survivors must manually complete a complex 12-page Department of Justice Form 06 affidavit in formal English, causing massive case drop-off before an Interim Protection Order is granted.

2. User Journey Story: Nomsa's Night in Soweto

Most safety apps only cover the 10 seconds of panic. Khusela covers Before, During, and After.

"It is 21:40 on a Thursday. Nomsa is in the kitchen. Her partner is in the next room, and he has checked her phone twice this week. She cannot make a phone call. If she photographs the bruise on her arm, it lands in her camera roll where he will see it. And tomorrow morning at the Magistrates Court, she will face a 12-page legal affidavit with nothing she can prove."

STEP 1: THE DISGUISE

Mzansi Smart-Budget

On her home screen, the app is a working ZAR & 15% VAT grocery calculator. If her partner grabs the phone, he sees bread (R18.50) and milk (R34.99). Zero suspicion.

STEP 2: SILENT ALARM

Dial 9999= in Plain Sight

She types 9999= on the calculator. The screen displays R160.49, while silently locking her GPS coordinates and preparing an emergency voice + SMS/WhatsApp dispatch.

STEP 3: THE VAULT

Dial 2580= for Proof

When safe, she types 2580=. She captures an off-gallery WebP photo and voice note. Each is encrypted with AES-256-GCM and sealed with an ECTA Section 15 SHA-256 hash.

STEP 4: THE COURT

1-Tap Form 06 (J480) PDF

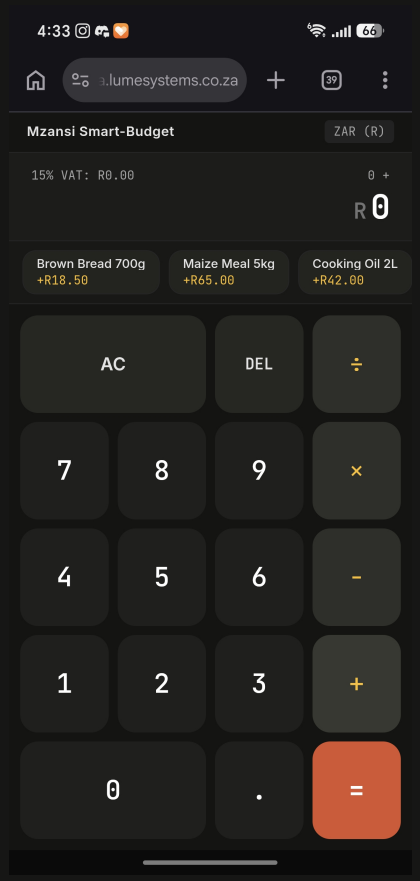
Her raw notes are structured into the official 12-page DoJ Form 06 + Form 6A + Annexure A SHA-256 manifest. If he walks in, Volume-Down snaps back to the calculator in <16ms.

3. Live Mobile Walkthrough (Part 1): Disguise to AI Affidavit

Real screenshots captured on mobile at https://khusela.lumesystems.co.za

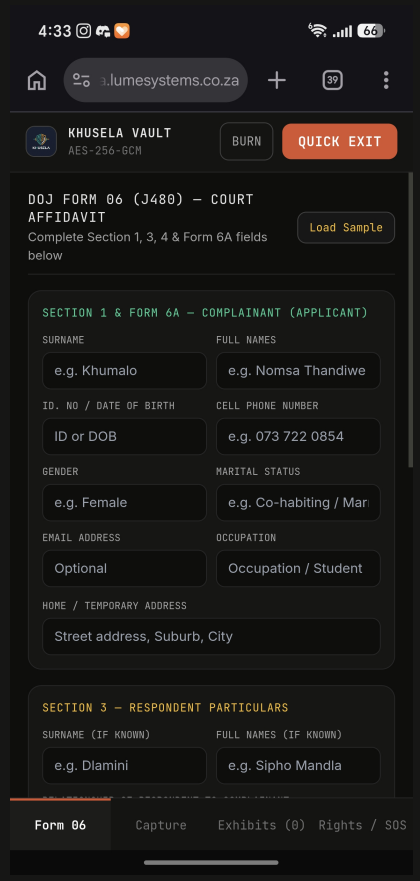
1. Calculator Disguise

2580= Vault | 9999= SOS



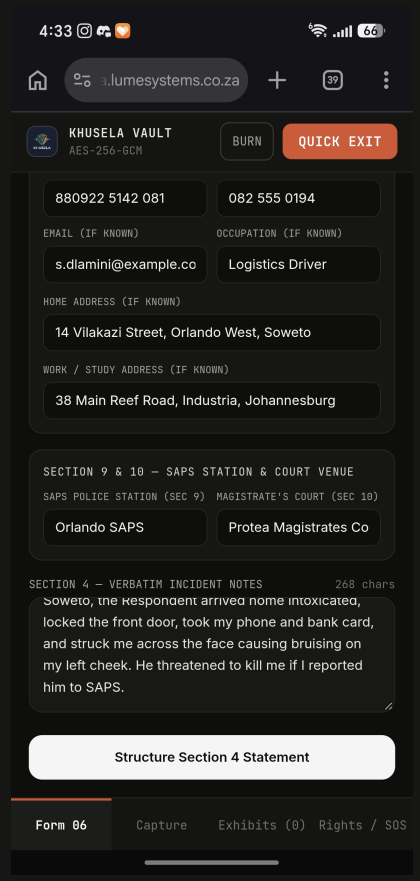
2. Unlocked Vault

AES-256-GCM Form 06



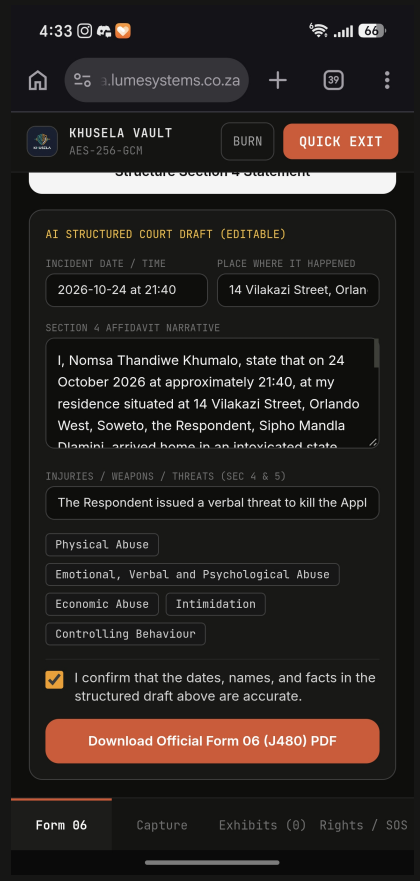
3. Verbatim Notes

SAPS & Court Venue



4. Gemini 3 Flash

Human-Verified Draft

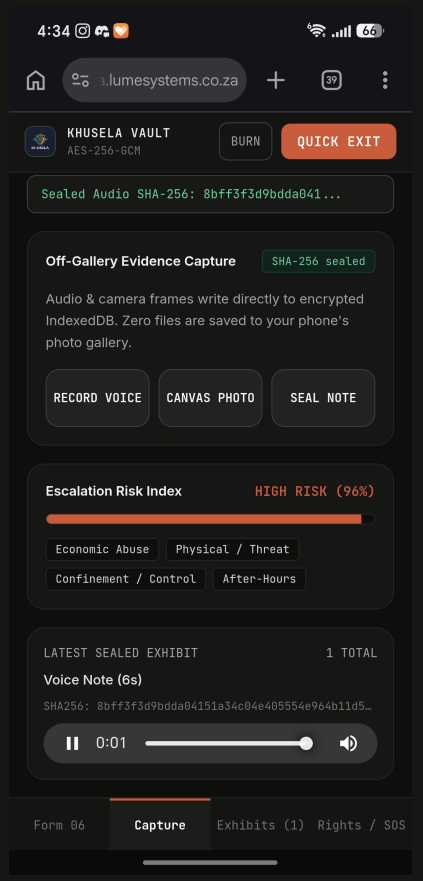


4. Live Mobile Walkthrough (Part 2): Forensic Vault & SOS Relay

Off-gallery WebP & voice capture, ECTA Section 15 SHA-256 sealing, live GPS pin, and WhatsApp relay.

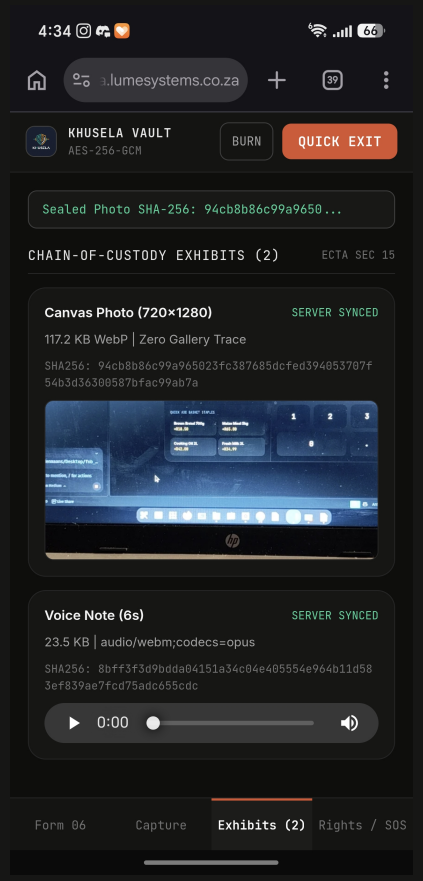
5. Off-Gallery Capture

87%-96% Risk Index



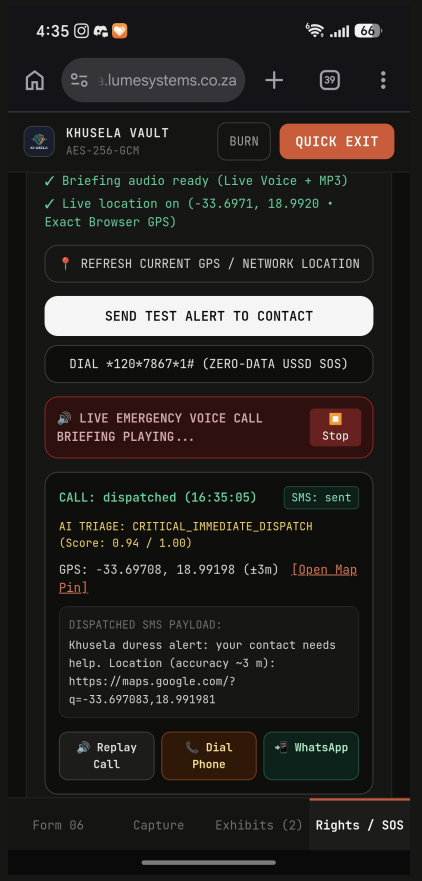
6. ECTA Sec 15 Seal

SHA-256 Chain of Custody



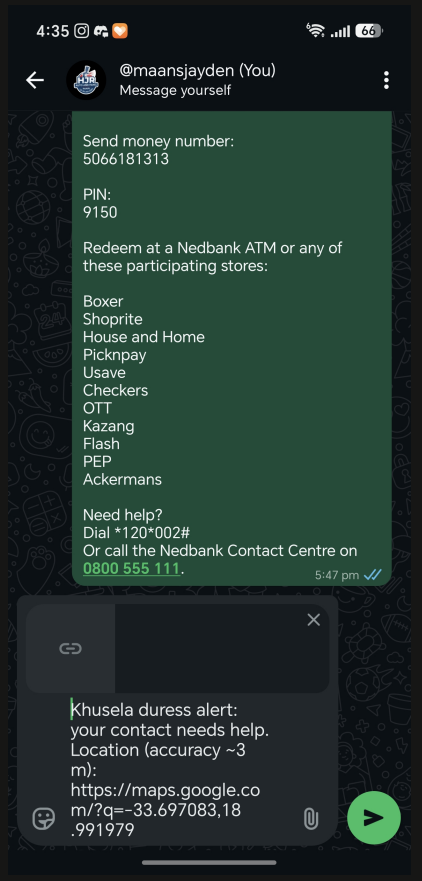
7. Live Voice & GPS

Exact GPS (±3m) + Triage



8. WhatsApp SOS Relay

Google Maps Pin Sent

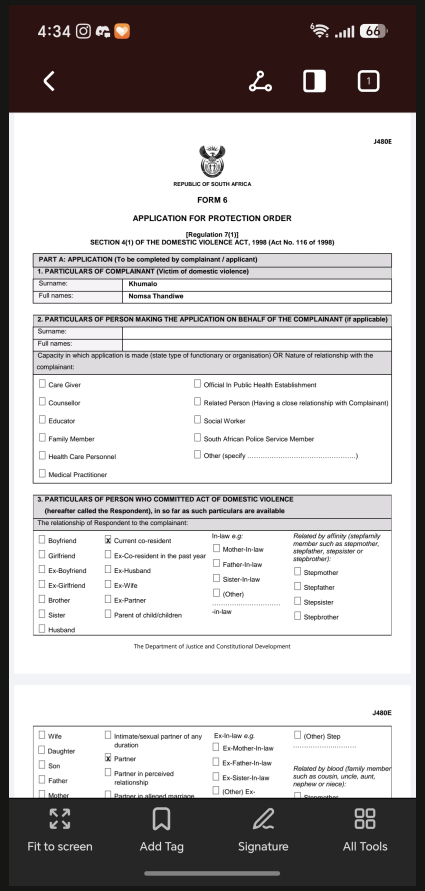


5. Official Completed DoJ Form 06 (J480) & Annexure A PDF

Actual pages from the 13-page Protection Order PDF generated in-memory on mobile (Zero disk caching).

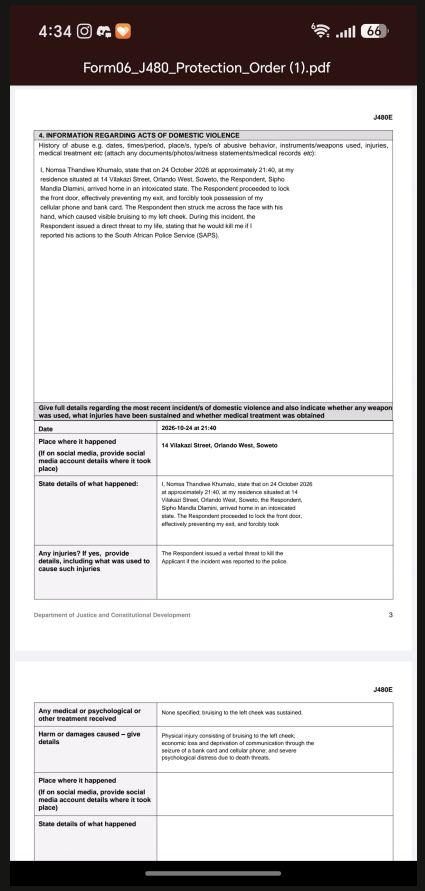
Mobile PDF View (Page 1)

Republic of SA Form 6 Header



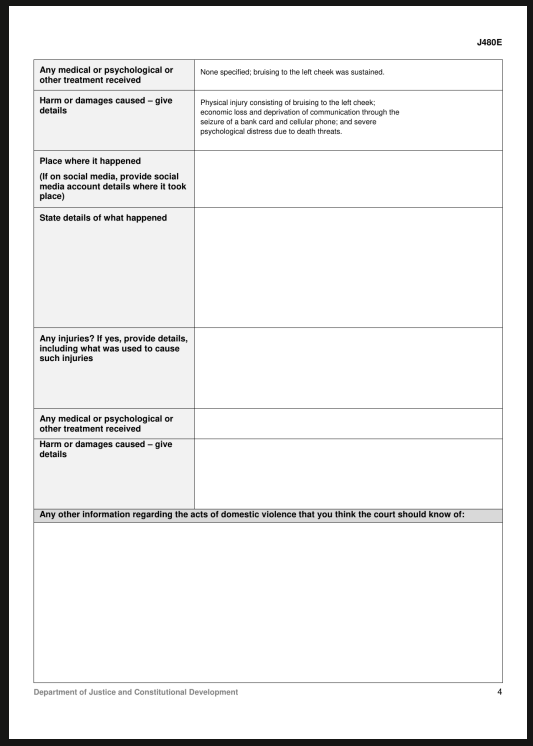
Mobile PDF View (Page 5)

Section 4 Structured Narrative



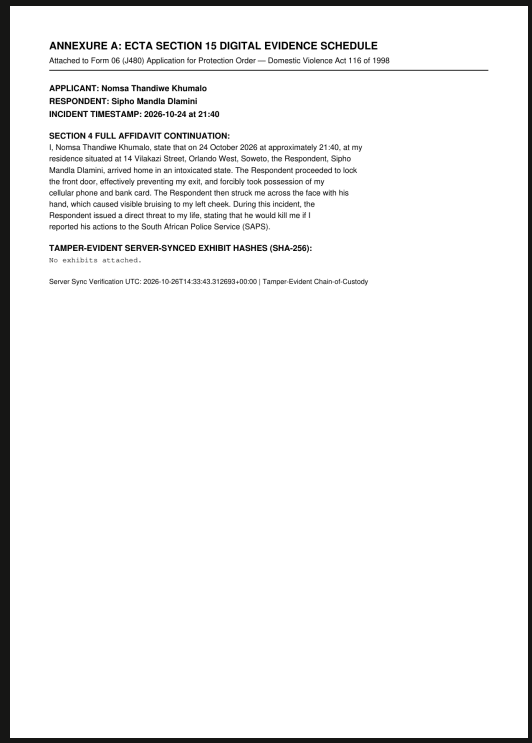
Page 4: Section 1 & 3

Complainant & Respondent Data



Page 13: Annexure A

ECTA Sec 15 SHA-256 Schedule



6. Technical Architecture: Zero-Knowledge Client + Stateless API

Single-origin Docker deployment on Render (HTTPS) serving the React 18 PWA and FastAPI backend.

1. CLIENT PWA (REACT 18)

- **Disguised Calculator UI (/)**
- **Zero URL / History leaks**
- **Web Crypto API (crypto.js):**
  - PBKDF2-SHA256 (100,000 iter)
  - AES-256-GCM Vault Storage
  - SHA-256 Exhibit Hashing
- **Off-Gallery <canvas> WebP**
- **Screen Wake Lock API**
- **Vol-Down / Shake Panic Exit**

2. FASTAPI BACKEND

- **POST /api/v1/sos/trigger**
  - Emergency Number Safety Lock
  - Call-First / Enrich-Second
- **GET /api/v1/sos/locate**
  - X-Forwarded-For SA IP Geo
- **POST /api/v1/evidence/sync**
  - Stateless SHA-256 & UTC Seal
- **POST /api/v1/legal/structure**
- **POST /api/v1/legal/generate-form06**

3. AI, PDF & TELEPHONY

- **Google Gemini 3 Flash**
  - Strict JSON schema output
  - Deterministic offline fallback
- **pypdf + ReportLab Engine**
  - In-memory BytesIO overlay
  - Official 12p DoJ Form 06 + 6A
  - Page 13 Annexure A Manifest
- **Twilio Voice/SMS + Free Relay**
  - WebRTC Voice + WhatsApp

7. SSDLC: Security Requirements, Threat Modelling & Testing

Engineered for a hostile domestic threat model where the adversary has physical access to the unlocked phone.

1. Security Requirements

POPIA & ECTA Section 15 compliance; PBKDF2-SHA256 (100k iterations) + AES-256-GCM client-side encryption; zero server-side storage of survivor PII or raw media.

2. Threat Modelling (STRIDE)

Coercive phone inspection mitigated via Calculator disguise + <16ms Quick Exit; Brute-force mitigated via 3-Strike Auto-Purge; PDF caching prevented via BytesIO + Cache-Control: no-store.

3. Secure Coding Practices

Strict Pydantic v2 bounds validation (-90..90 lat, -180..180 lng); html.escape() XML sanitization on all TwiML inputs; zero hardcoded secrets; path-traversal confinement via Path.resolve().

4. Automated Security Testing

Verified via backend/test_api.py: tests emergency shortcode blocklist (10111, 10177, 112, 911, 0800428428), honest failure reporting, SHA-256 tamper detection, and /..%2f path traversal guard.

5. Secure Integration

Twilio REST API restricted to verified contact; Gemini 3 Flash isolated from raw vault keys with mandatory human confirmation; Docker container runs over TLS 1.3 on Render.

8. Sustainability, Corporate Win-Win & Funding Plan to 100%

100% Free Forever for Survivors | Transparent Hackathon Twilio Trial Limitation & Immediate Production Plan

LEAN BUSINESS CANVAS: TAX & B-BBEE WIN-WIN

R0.00 Forever for Survivors

Charging a survivor creates a dangerous bank statement trail. Core safety, vault, and Form 06 tools are 100% free.

1. Section 18A Tax Write-Off + B-BBEE SED

Corporate sponsors (banks, telcos, insurers) receive SARS Section 18A tax deductions, verified B-BBEE Socio-Economic Development points, and audited ESG impact metrics.

2. Private Security API Licensing (R4,500/mo)

Armed-response partners integrate our Call-First webhook and AI Threat Triage feed into their dispatch consoles.

3. Legal Aid & NGO Case Portals (R1,500/mo)

Bulk court-ready Form 06 (J480) + Annexure A intake for legal clinics, against a lean R6,500/month cloud & telecom operating cost.

TRANSPARENT TWILIO LIMITATION & FUNDING PLAN

Why Silent 9999= Carrier Calls Are on Failover Today: Our hackathon Twilio trial account has a USD 0.00 balance and no purchased phone number, so Twilio blocks live PSTN carrier calls. In the demo, Khusela automatically fails over to our free WebRTC Voice Briefing + SMS preview + 1-tap WhatsApp relay.

Phase 1 (Week 1 - ~R1,000):
Upgrade Twilio account & provision a South African voice/SMS number. Activates live 9999= carrier calls & SMS with zero code changes.

Phase 2 (Week 2 - R350/mo):
Always-on production container (zero cold-start sleep) + 24/7 uptime monitoring.

Phase 3 (Month 1-2):
Live USSD (*120*7867#) aggregator integration for zero-data feature phones + multi-contact escalation.

Phase 4 (Month 2-3):
Private security control-room webhook dispatch & POPIA/independent security audit.

9. Test Khusela Live in 60 Seconds (Judge's Guide)

No installation or compiling needed. Works immediately in any mobile or desktop browser.

INTERACTIVE JUDGE WALKTHROUGH

1. Open the Live App

Visit https://khusela.lumesystems.co.za on your phone or PC. You will see Mzansi Smart-Budget, a working ZAR & 15% VAT calculator.

2. Unlock the Encrypted Vault (PIN: 2580=)

Type 2580 and press =. The hidden AES-256-GCM Khusela Vault opens immediately.

3. Generate an Official DoJ Form 06 (J480) PDF

On the Form 06 tab, click 'Load Sample', click 'Structure Section 4 Statement', tick the confirmation checkbox, and click 'Download Official Form 06 (J480) PDF' to inspect all 13 pages.

4. Test Forensic Capture & SHA-256 Sealing

Open the Capture tab to record a voice note, canvas photo, or incident note, then inspect its SHA-256 digest in Exhibits.

5. Test Multilingual Rights & SOS Alert

Open Rights / SOS, play Domestic Violence Act rights in English, Afrikaans, isiXhosa, or isiZulu, and click 'SEND TEST ALERT TO CONTACT'.

KHUSELA

Quick Reference Card

Web App URL:	**khusela.lumesystems.co.za**
Vault Unlock PIN:	**2580=**
Silent SOS PIN:	**9999=**
Panic Exit Key:	**QUICK EXIT / Vol-Down / Esc**
3-Strike Wipe:	**3x wrong 4-digit PINs**
Demo Video:	**Click Here (Google Drive)**
GitHub Repo:	**maansjayden/khusela-safeledger**